

from higher school physics, you will have seen superposition in action. It is this behaviour, of being in both states at once, that can be harnessed to perform almost miraculous feats of computing—such as examining both sides of a coin at once, investigating simultaneously the calculations of both a positive and negative input, and instantly searching vast databases.

A bit, represented as a quantum mechanical particle, is called a Quantum Bit, or qubit for short. Any property of a quantum mechanical particle can theoretically be used to store a qubit, and—in these early days—researchers are experimenting with many different kinds of materials to hold qubits. Electron ‘spin’ and polarization of photons seem to be the most popular media.

A photon is a particle of light. Polarization of a photon is simply the plane along which the light wave vibrates. Wave or particle? It’s both! Spin is a property that is very hard to describe, and is usually just abstracted as the direction of

demonstrated last year that entanglement works even when one of the particles is fired through a sheet of metal, thick enough to block light. Albert Einstein himself described it as “spooky action at a distance.” We are in good company.

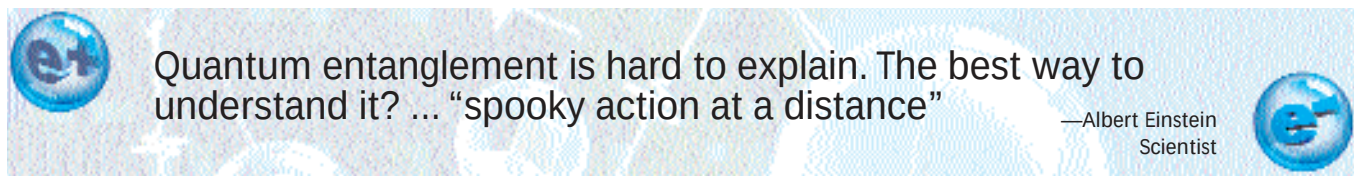
Boundless opportunities

It doesn’t take much effort to figure out that such a means of communication opens up huge opportunities, including wireless circuitry in quantum computers and teleportation sometime in the next century. Scientists from the University of Vienna in Austria have indeed been able to teleport qubits around over a stable link of quantum entangled photons. Using entanglement to ‘wire’ qubits together is essential in manufacturing a quantum computer. A discovery early this year in the University of Michigan heralds a promising new technique to entangle three or more electrons to communicate ‘spin’.

According to the University of Michi-

two states at once, with instantaneous connections through entanglement, and you have the makings of a quantum computer. One that can evaluate the probability of a combined set of states of hundreds of thousands of qubits in a single step, and produce the result instantly. In that instant of observation, the qubit array could have been in all of billions of possible states. Yet at the end it must choose one, and that is the result of the calculation. The killer application for parallel computing at this unparalleled scale is Grover’s Search Algorithm—a quantum algorithm to search across an unordered list.

A traditional computer must examine each item in the list for a match, in serial order. A computer with 10 CPUs could do this in 1/10th the time. A quantum computer, on the other hand, can—with enough qubits—examine thousands of items at once, faster than yet possible. Similarly quantum algorithms for factoring large numbers are expected to create havoc in cryptography. Encryption algo-



rotation of a particle. There is more to it, though, since you may come across particles with $1\frac{1}{2}$ spin as well. Enough to make your head spin!

Fit to be tied

Not satisfied with all the funny business about determining the state of the qubit, we must also deal with quantum entanglement. If you were baffled before, please brace yourselves for this one. Two quantum particles, a pair of photons or electrons for example, can be ‘entangled’, such that any change in one particle is instantly reflected in the other.

Here is a simple thought experiment you can try at home. Hold on to a mental picture of two particles in close contact, transmitting their state to each other. Now take them 10nm apart. Change the state of one, and the other undergoes the same change. Now take them 10 metres apart, repeat process and get the same result. Now take them 10 kilometres apart. Repeat. 10 light-years apart. Repeat. No matter the distance, entangled particles are soul mates like no other. Scientists in a Dutch University

gan News Service, here is how they did it—“In the presence of an external magnetic field, a resonant laser pulse creates localised excitons (bound electron-hole pairs) of radius ~ 0.005 microns in the CdTe well. Electrons bound to donor impurities within that radius, feel the presence of the exciton in such a way that they became entangled after the exciton is gone. The process involves resonant Raman transitions between Zeeman split spin states. In the experiments, the signature of entanglement involving ‘m’ electrons is the detection of the ‘mth’-harmonic of the fundamental Zeeman frequency in the differential reflectivity data.” Interesting.

Calculation

The strength of quantum computers is not that they can take a stream of calculations and process them in serial order at high speeds—like the hot P4 or Athlon XP that powers your desktop computer. They are massively parallel computation devices, and if you are thinking of supercomputers with thousands of CPUs, think again.

Combine the ability of a qubit to be in

rithms hide behind the fact that there isn’t enough computational power to crack the code, especially with larger keys. Quantum Computers will be able to rip through them with ease...but don’t expect the Model3000ZX Quantum Crack-o-matic anytime soon.

Quantum cryptography

One of the fundamental problems in cryptography, is how do you get the encryption key (or password, if you prefer) to the receiver, secretly. There is currently no way to guarantee that a snooper has not read the key on the way to the receiver. One workaround is Public Key Cryptography, that involves a shared public key which the whole world knows, and a hidden private key that only you can use. As explained earlier, with quantum computers, it will be possible to crack even large keys. It will also be possible to infer the private key for a given public key. No need to panic already, though, since quantum computers aren’t expected anytime soon.

Yet another fundamental and counter-intuitive property of quantum particles is the slipperiness. According to Heisenberg’s